Hypothesis Testing and Decision Errors

Lectures 20–21

QBA201 – Quantitative Business Analysis

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Would you investigate the filling machine?

Start from the last lesson

An interval estimates the population mean. Today the factory already has a target: **500 mL**.

A sample of bottles averages **503 mL**. How much could sample averages vary if the machine were on target?

1. Write the claim about the process.
2. Describe what sample means look like under that claim.
3. Use the relevant tail area to make the decision.

Write the claim before reading the data

Let μ be the true mean fill across bottles from the process.

$$H_0: \ \mu = 500 \text{ mL} \qquad H_1: \ \mu \neq 500 \text{ mL}$$

- Underfilling and overfilling both matter, so the question is two-sided.
- $\bar{x} = 503$ mL is the observed *sample* mean.

If H_0 were true, how unusual would a sample mean this far from 500 be?

The alternative determines the direction

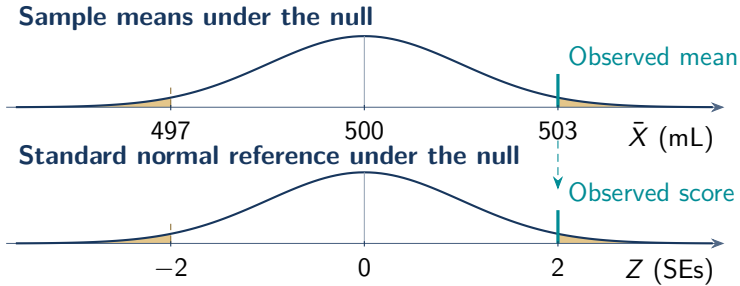
Test	Null	Alternative
Two-sided	$\mu = c$	$\mu \neq c$
Left-tailed	$\mu \geq c$	$\mu < c$
Right-tailed	$\mu \leq c$	$\mu > c$

For one-sided tests, calculate the reference distribution at boundary $\mu = c$.

Discuss: Why should the direction be chosen before seeing the sample mean?

One null model on two scales

Independent normal fills, $n = 64$, known $\sigma = 12$ mL. Under H_0, the mean of $\bar{X}$ is 500 mL and its standard error is $12/\sqrt{64} = 1.5$ mL.



$$z_{\text{obs}} = \frac{503 - 500}{1.5} = 2 \qquad \underbrace{\mathrm{P}_{H_0}(|\bar{X} - 500| \geq 3)}_{\text{mL scale}} = \underbrace{\mathrm{P}(|Z| \geq 2)}_{\text{standardized scale}} \approx 0.0455$$

Compare the tail area with the false-alarm limit

- Before sampling, set $\alpha = 0.05$. The rule may trigger a false alarm in 5% of repeated samples if H_0 is true.
- Our observed $z = 2$ gives $p = 0.0455$: about 4.55% of on-target samples would be at least this far from 500 mL.
- Since $p < \alpha$, **reject** H_0. There is evidence that the process mean differs from 500 mL.

A larger p-value would leave the evidence inconclusive. The size of the fill difference and the cost of an adjustment are separate business questions.

Three ways to make the same decision

Two-sided 5% test of $H_0 : \mu = 500$ mL, with SE $= 1.5$ mL.

Scale	Compare	Result
Sample mean	$503 > 500 + 1.96(1.5) = 502.94$	Outside the range
Standardized score	$\lvert 2 \rvert > 1.96$	Beyond the cutoff
Tail probability	$0.0455 < 0.05$	Below the limit

All three reject H_0 using the same test. The p-value is a probability of sample results calculated under H_0.

The known-SD two-sided rule

$$z_{\text{obs}} = \frac{\bar{x} - c}{\sigma/\sqrt{n}}.$$

Reject if:

$$|z_{\text{obs}}| \geq z_{\alpha/2}.$$

Equivalently, reject if the corresponding $(1 - \alpha)$ confidence interval excludes c.

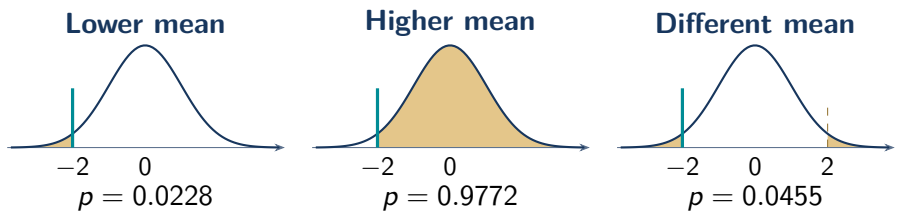
One-sided rules keep the specified direction

Alternative	Rejection rule	At $\alpha = .05$
$\mu < c$	$z_{\text{obs}} \leq -z_\alpha$	$z_{\text{obs}} \leq -1.6449$
$\mu > c$	$z_{\text{obs}} \geq z_\alpha$	$z_{\text{obs}} \geq 1.6449$

Discuss: Why does using $|z|$ lose information needed for a one-sided decision?

The alternative tells you which area to count

Compare three *preplanned* questions at the same observed $z = -2$.



$\Phi(z_{\text{obs}})$ (left) $\quad 1 - \Phi(z_{\text{obs}})$ (right) $\quad 2[1 - \Phi(|z_{\text{obs}}|)]$ (two-sided)

Pick the tail from the business question before looking at the sample.

Two error types refer to the true population state

Decision	H_0 true	H_0 false
Reject	Type I error	Correct detection
Fail to reject	Correct nonrejection	Type II error

α controls Type I error under the null.

β is Type II probability at a specified alternative. Power is $1 - \beta$.

A rejection rule with a known error probability

Assume normal fill volumes with known $\sigma = 12$ ml and $n = 64$.

$$H_0 : \mu = 500, \qquad H_1 : \mu \neq 500.$$

Rule: reject when $|\bar{X} - 500| \geq 3$ ml.

Discuss: Calculate the probability of rejection when H_0 is true.

Complete the Type I calculation

$$\mathrm{SE} = 12/\sqrt{64} = 1.5, \qquad 3/\mathrm{SE} = 2.$$

$$\alpha = \mathrm{P}(|Z| \geq 2) = 2[1 - \Phi(2)] = 0.0455.$$

The prescribed rule has Type I probability about 4.55%.

A two-sided decision at ten-percent significance

For $H_0 : \mu = 0$ versus $H_1 : \mu \neq 0$, suppose $z_{\text{obs}} = 1.7$.

$$|1.7| > 1.6449.$$

Reject at $\alpha = .10$.

$$p = 2[1 - \Phi(1.7)] = 0.0891.$$

The p-value leads to the same decision.

Keep one-tail and two-tail critical values distinct

α	One-tail z_α	Two-tail $z_{\alpha/2}$
.10	1.2816	1.6449
.05	1.6449	1.9600
.01	2.3263	2.5758

Discuss: Why does 1.6449 appear in two different places?

Constructed salary example: a two-sided question

Suppose annual salary observations have known $\sigma = 30,000$, $n = 100$, and $\bar{x} = 185,400$.

$$H_0 : \mu = 180,000, \quad H_1 : \mu \neq 180,000, \quad \alpha = .10.$$

$$\text{SE} = 3,000, \quad z = 1.8, \quad p = 0.0719.$$

Reject at 10%, but not at 5%.

Constructed salary example: a left-tail question

Use $n = 100$, known $\sigma = 30,000$, and $\bar{x} = 174,600$.

$$H_0 : \mu \geq 180,000, \qquad H_1 : \mu < 180,000.$$

$$z = -1.8, \qquad p = \Phi(-1.8) = 0.0359.$$

Reject at $\alpha = .05$. The data provide evidence that the mean is below 180,000 under this model.

A mean claim is different from a rarity claim

If the claim is that *mean* salary is at most 200,000:

$$H_0 : \mu \leq 200,000, \quad H_1 : \mu > 200,000.$$

For constructed $\bar{x} = 205,400$, $n = 100$, $\sigma = 30,000$:

$$z = 1.8, \quad p = 0.0359.$$

Reject at 5%: evidence supports a population mean above 200,000.

A claim that individuals rarely exceed 200,000 concerns a tail proportion, not simply the mean.

The question determines the standard error

Let d_0 be the null difference. For every row, use $z = (\text{estimate} - \text{null value})/\text{null standard error}$.

Target	Estimate and null value	Standard error for the test
One mean, known σ	$\bar{x},\ \mu_0$	$\sigma/\sqrt{n}$
One proportion	$\hat{p},\ p_0$	$\sqrt{p_0(1-p_0)/n}$
Difference in means	$\bar{x}_1 - \bar{x}_2,\ d_0$	$\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$
Difference in proportions	$\hat{p}_1 - \hat{p}_2,\ 0$	$\sqrt{\hat{p}_{\text{pool}}(1-\hat{p}_{\text{pool}})(1/n_1 + 1/n_2)}$

For the difference rows, use independent groups. Mean SDs are known. In the equality test of two rates, $\hat{p}_{\text{pool}}$ is total successes divided by total observations.

Type II probability depends on the alternative

$$\beta(\mu_1) = \mathrm{P}_{\mu=\mu_1}(\text{fail to reject } H_0).$$

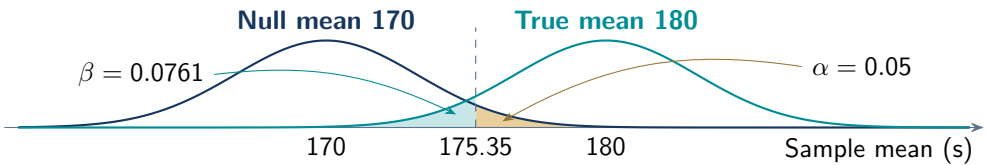
Specify the rejection rule, sample size, variability and true alternative value before calculating.

Different alternative means generally give different β values.

Power is a function of the alternative, not a universal property of a test name.

One cutoff, two possible process means

Billing time example. Test whether the population mean exceeds 170 seconds. The standard error of the sample mean is 3.25 seconds.



Cutoff $= 170 + 1.6449(3.25) \approx 175.35$ Power at $180 = 1 - 0.0761 = 0.9239$

A stricter alpha lowers power at the same alternative

Keep $\mu_0 = 170$, true mean 180 and SE $= 3.25$.

$$\alpha = .01: \quad c = 170 + 2.326348(3.25) = 177.5606.$$

$$\beta = \Phi\left(\frac{177.5606 - 180}{3.25}\right) = 0.2265.$$

Power is 0.7735.

Discuss: Why does moving the rejection cutoff upward make detection harder?

More observations can improve the tradeoff

At fixed population SD:

$$\text{SE} = \sigma/\sqrt{n}.$$

A larger valid sample reduces SE and can increase power at a fixed α for a fixed nonnull alternative.

Better sampling and measurement remain essential. More biased observations are not a substitute for a credible design.